

**2025-2026 Fall**  
**MAT123 Final**  
**09/01/2026**

1. a. Two vertices of a triangle lie on the curve  $y = \sqrt{4 - x^2}$  and one vertex is at the origin. One edge is parallel to the  $x$ -axis. What is the maximum area of the triangle?

b. i. Evaluate  $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_x^{\pi/2} \ln(\sin t) dt}{\sin x - 1}$ .

ii. Evaluate the sum  $\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{2 + \frac{3i}{n}}$ . (*Hint: Riemann sum*)

2. a. Set up the integral (but do not evaluate) that gives the area under the line  $\sqrt{7} \cdot y = x + 1$ , left to the line  $x = \frac{3}{4}$ , and inside the circle  $x^2 + y^2 = 1$ .

b. i. Evaluate  $\int \frac{\cos^3 x + \cos x}{\sin^3 x - \sin^2 x} dx$ .

ii. Evaluate  $\int_{-\pi}^{\pi} \left(1 + \sin^3 x \cdot \sqrt{9x^2 + 4 \cos^2 x}\right) dx$ .

3. a. Determine if the improper integral  $\int_0^3 \frac{(1 + e^{-x})(x^3 + x + 1)}{x^5} dx$  converges or diverges with the Comparison Test. You cannot use the Limit Comparison Test.

- b. i. The region  $R$  is bounded by  $y = x^3 + x$ ,  $y = 2$ , and  $x = 0$ . Find the volume of the object if  $R$  is rotated around the  $x$ -axis.

- ii. The region  $R$  is bounded by  $y = x^3 + x$ ,  $y = 2$ , and  $x = 0$ . Find the volume of the object if  $R$  is rotated around the  $y$ -axis.

4. a. Determine the radius and interval of convergence of  $\sum_{n=0}^{\infty} \frac{(-1)^n (x - 3)^n}{2^n \cdot n}$ .

b. i. Is  $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$  convergent or divergent?

ii. Is  $\sum_{n=1}^{\infty} \frac{1}{3^{n \sin(\frac{1}{n})}}$  convergent or divergent?

1. a. Let  $0 \leq x \leq 2$ . The base of the triangle parallel to the  $x$ -axis has length  $2x$ , and its height is  $y = \sqrt{4 - x^2}$ . The area function is

$$A(x) = \frac{2x \cdot y}{2} = x\sqrt{4 - x^2}.$$

Differentiating with respect to  $x$ ,

$$\frac{dA}{dx} = \sqrt{4 - x^2} + x \cdot \frac{-2x}{2\sqrt{4 - x^2}} = \frac{4 - 2x^2}{\sqrt{4 - x^2}}.$$

Setting  $\frac{dA}{dx} = 0$  gives  $4 - 2x^2 = 0 \implies x = \sqrt{2}$ .

Evaluating the area at  $x = \sqrt{2}$ ,

$$A(\sqrt{2}) = \sqrt{2}\sqrt{4 - (\sqrt{2})^2} = \sqrt{2} \cdot \sqrt{2} = \boxed{2}.$$

- b. i. Evaluating the limit using L'Hôpital's Rule (form  $\left[\frac{0}{0}\right]$ ):

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\int_x^{\pi/2} \ln(\sin t) dt}{\sin x - 1} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\ln(\sin x)}{\cos x}$$

Applying L'Hôpital's Rule again (form  $\left[\frac{0}{0}\right]$ ):

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{-\frac{\cos x}{\sin x}}{-\sin x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos x}{\sin^2 x} = \frac{0}{1} = \boxed{0}.$$

- ii. Using the definite integral definition of a Riemann sum:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{b-a}{n} f\left(a + i \frac{b-a}{n}\right) = \int_a^b f(x) dx$$

Here,  $a = 2$ ,  $b = 5$ ,  $\frac{b-a}{n} = \frac{3}{n}$ , and  $f(x) = \sqrt{x}$ . Therefore:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{3}{n} \sqrt{2 + \frac{3i}{n}} = \int_2^5 \sqrt{x} dx = \left[ \frac{2}{3} x^{3/2} \right]_2^5 = \boxed{\frac{2}{3} (5\sqrt{5} - 2\sqrt{2})}$$

2. a. To find the intersection points of the line  $\sqrt{7}y = x + 1$  and the circle  $x^2 + y^2 = 1$ :

$$y^2 = \frac{(x+1)^2}{7} \implies x^2 + \frac{x^2 + 2x + 1}{7} = 1 \implies 8x^2 + 2x - 6 = 0 \implies 4x^2 + x - 3 = 0$$

Solving  $(4x - 3)(x + 1) = 0$  gives  $x = -1$  and  $x = \frac{3}{4}$ .

The area under the line and inside the circle is given by:

$$A = \int_{-1}^{3/4} \left( \frac{x+1}{\sqrt{7}} - (-\sqrt{1-x^2}) \right) dx = \boxed{\int_{-1}^{3/4} \left( \frac{x+1}{\sqrt{7}} + \sqrt{1-x^2} \right) dx}$$

- b. i. Let  $I = \int \frac{\cos^3 x + \cos x}{\sin^3 x - \sin^2 x} dx = \int \frac{(\cos^2 x + 1) \cos x}{\sin^2 x (\sin x - 1)} dx$ . Substitute  $u = \sin x$ , so  $du = \cos x dx$  and  $\cos^2 x = 1 - u^2$ .

$$I = \int \frac{2 - u^2}{u^2(u - 1)} du.$$

Using partial fraction decomposition:

$$\frac{2 - u^2}{u^2(u - 1)} = \frac{A}{u} + \frac{B}{u^2} + \frac{C}{u - 1} \implies Au(u - 1) + B(u - 1) + Cu^2 = 2 - u^2$$

Setting  $u = 0 \implies -B = 2 \implies B = -2$ .

Setting  $u = 1 \implies C = 1$ .

Comparing  $u^2$  coefficients:  $A + C = -1 \implies A + 1 = -1 \implies A = -2$ .

Integrating each term:

$$I = \int \left( \frac{-2}{u} - \frac{2}{u^2} + \frac{1}{u - 1} \right) du = -2 \ln |u| + \frac{2}{u} + \ln |u - 1| + C$$

Substituting back  $u = \sin x$ :

$$I = -2 \ln |\sin x| + \frac{2}{\sin x} + \ln |\sin x - 1| + C$$

- ii. Split the integral into two parts:

$$\int_{-\pi}^{\pi} \left( 1 + \sin^3 x \sqrt{9x^2 + 4 \cos^2 x} \right) dx = \int_{-\pi}^{\pi} dx + \int_{-\pi}^{\pi} \sin^3 x \sqrt{9x^2 + 4 \cos^2 x} dx$$

The integrand  $f(x) = \sin^3 x \sqrt{9x^2 + 4 \cos^2 x}$  is an odd function because  $f(-x) = -\sin^3 x \sqrt{9(-x)^2 + 4 \cos^2(-x)} = -f(x)$ . Since the interval  $[-\pi, \pi]$  is symmetric about 0, the integral of the odd function vanishes.

$$= [x]_{-\pi}^{\pi} + 0 = \pi - (-\pi) = \boxed{2\pi}$$

3. a. For  $x \in (0, 3]$ , we have  $e^{-x} > 0 \implies 1 + e^{-x} > 1$ , and  $x^3 + x + 1 > 1$ . Therefore,

$$\frac{(1 + e^{-x})(x^3 + x + 1)}{x^5} \geq \frac{1}{x^5}.$$

Consider the integral  $I = \int_0^3 \frac{1}{x^5} dx$ .

$$I = \int_0^3 x^{-5} dx = \lim_{R \rightarrow 0^+} \left[ \frac{x^{-4}}{-4} \right]_R^3 = \lim_{R \rightarrow 0^+} \left( -\frac{1}{4(3^4)} + \frac{1}{4R^4} \right) = \infty$$

Since  $I$  diverges and  $\frac{(1+e^{-x})(x^3+x+1)}{x^5} \geq \frac{1}{x^5} > 0$  on  $(0, 3]$ , by the Direct Comparison Test, the integral  $\int_0^3 \frac{(1+e^{-x})(x^3+x+1)}{x^5} dx$  also diverges.

- b. i. Rotating the region  $R$  bounded by  $y = x^3 + x$ ,  $y = 2$ , and  $x = 0$  around the  $x$ -axis using the Washer Method ( $x \in [0, 1]$ ):

$$V_1 = \pi \int_0^1 ((2)^2 - (x^3 + x)^2) dx = \pi \int_0^1 (4 - (x^6 + 2x^4 + x^2)) dx$$

$$V_1 = \pi \left[ 4x - \frac{x^7}{7} - \frac{2x^5}{5} - \frac{x^3}{3} \right]_0^1 = \pi \left( 4 - \frac{1}{7} - \frac{2}{5} - \frac{1}{3} \right) = \boxed{\frac{328\pi}{105}}$$

- ii. Rotating the same region around the  $y$ -axis using the Shell Method.

$$V_2 = 2\pi \int_0^1 x (2 - (x^3 + x)) dx = 2\pi \int_0^1 (2x - x^4 - x^2) dx$$

$$V_2 = 2\pi \left[ x^2 - \frac{x^5}{5} - \frac{x^3}{3} \right]_0^1 = 2\pi \left( 1 - \frac{1}{5} - \frac{1}{3} \right) = 2\pi \left( \frac{7}{15} \right) = \boxed{\frac{14\pi}{15}}$$

4. a. Using the Ratio Test for  $\sum_{n=1}^{\infty} \frac{(-1)^n (x-3)^n}{2^n \cdot n}$ :

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-3)^{n+1}}{2^{n+1}(n+1)} \cdot \frac{2^n \cdot n}{(x-3)^n} \right| = \frac{|x-3|}{2} \lim_{n \rightarrow \infty} \frac{n}{n+1} = \frac{|x-3|}{2}$$

For convergence,  $L < 1 \implies \frac{|x-3|}{2} < 1 \implies |x-3| < 2$ . Thus, the radius of convergence is  $\boxed{R=2}$ .

Solving  $|x-3| < 2$  yields  $1 < x < 5$ .

At  $x = 1$ :  $\sum_{n=1}^{\infty} \frac{(-1)^n (-2)^n}{2^n \cdot n} = \sum_{n=1}^{\infty} \frac{1}{n}$ , which is the harmonic series and diverges.

At  $x = 5$ :  $\sum_{n=1}^{\infty} \frac{(-1)^n 2^n}{2^n \cdot n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ , which converges by the Alternating Series Test.

Therefore, the interval of convergence is  $\boxed{(1, 5]}$ .

- b. i. Applying the Integral Test for  $f(x) = \frac{1}{x(\ln x)^2}$  on  $[2, \infty)$ :

$$\int_2^{\infty} \frac{1}{x(\ln x)^2} dx = \lim_{R \rightarrow \infty} \int_{\ln 2}^{\ln R} \frac{1}{u^2} du = \lim_{R \rightarrow \infty} \left[ -\frac{1}{u} \right]_{\ln 2}^{\ln R} = \frac{1}{\ln 2}.$$

Since the integral converges, the series  $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$   $\boxed{\text{converges}}$ .

- ii. Using the  $n$ -th Term Test for Divergence,

$$L = \lim_{n \rightarrow \infty} \frac{1}{3^{n \sin(1/n)}}$$

Let  $u = 1/n$ . As  $n \rightarrow \infty$ ,  $u \rightarrow 0^+$ .

$$\lim_{n \rightarrow \infty} n \sin(1/n) = \lim_{u \rightarrow 0^+} \frac{\sin u}{u} = 1 \implies \lim_{n \rightarrow \infty} \frac{1}{3^{n \sin(1/n)}} = \frac{1}{3^1} = \frac{1}{3} \neq 0$$

Since the limit is not 0, the series  $\boxed{\text{diverges}}$  by the  $n$ th Term Test.